# Coastal Representation Learning by Combining Optical and Depth Foundation Models

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## Abstract

Coastal prediction combines spectral appearance, spatial context and geometric structure, but these signals are rarely provided by one pretrained encoder. We study a simple dual-foundation architecture that combines an optical Earth-observation model, OLMoEarth Large, with the visual encoder of PromptDA, a depth foundation model. Both encoders remain frozen: their spatial tokens are aligned and concatenated before a task-specific linear probe. The evaluation uses four downstream tasks—bathymetry regression and three segmentation benchmarks—under a common validation-selected probing protocol. On 1,150 coastal regions used for coastal adaptation experiments, the dual-foundation representation improves MARIDA mIoU from 0.6129 to 0.6384 and MADOS mIoU from 0.6480 to 0.6833 relative to OLMoEarth Large alone. Bathymetry changes from 0.5771 to 0.5749 m RMSE, whereas LICS decreases from 0.9548 to 0.9431. We report OLMoEarth Nano adaptation and a task-adapted SpectralMoE implementation as controls that separate the dual-foundation result from additional coastal or target-task supervision. The results support optical–depth feature complementarity on selected coastal tasks, while the single-run design, larger concatenated probe and unmatched interfaces limit claims of general superiority.


## 1. Introduction

Coastal prediction spans tasks with different information requirements. Shoreline delineation depends on boundaries, marine surface interpretation depends on spectral and contextual cues, and bathymetry requires continuous geometric estimates. Optical Earth-observation models are well matched to spectral variation, but a depth foundation model can provide a complementary bias toward scene layout and relative structure. The central question is whether these two pretrained representations can be combined without retraining either encoder.

OLMoEarth provides a multimodal optical representation and a frozen-feature evaluation recipe [1]. PromptDA extends the Depth Anything line toward high-resolution metric-depth estimation [3]; here we use its visual encoder features, not its depth predictions or depth prompts. Combining the two models is attractive because it adds a geometric representation without requiring paired depth labels in the coastal collection. The combination also creates practical questions about spatial alignment, feature dimensionality and whether any gain survives a frozen, task-independent evaluation.

We investigate the dual-foundation design through a local evaluation study rather than new pretraining. We first establish ten pretrained-model references under a common probing budget. We then compare OLMoEarth Large alone with frozen OLMoEarth–PromptDA concatenation on four tasks. OLMoEarth Nano coastal adaptation and a separate transfer of the released SpectralMoE architecture [2] provide controls for the effects of coastal supervision and target-task adaptation. These controls are reported separately because they change the representation before linear probing.

Our contribution is an explicit and reproducible baseline for combining optical and depth foundation-model features in coastal remote sensing. The method has three deliberately small operations: independent preprocessing, spatial-token alignment and channel concatenation, followed by a task-specific probe. The

experiments show selected gains on marine-surface segmentation while exposing a trade-off on shoreline segmentation. We do not claim a new depth model, a new optical encoder or a new fusion primitive; the scientific claim is narrower: a frozen depth foundation model can complement an optical foundation model on some coastal transfers under the tested protocol.

# 2. Data and evaluation settings

## 2.1. Coastal adaptation collection

We use 1,150 coastal regions selected from an existing collection for complete monthly records in Sentinel-2, Landsat, and MODIS during 2024. The split contains 911 training, 120 validation, and 119 held-out regions. Months from a region retain the same split membership. Complete monthly records do not imply that every pixel is valid; observation masks remain necessary.

Sentinel-2 supplies 12 bands per time step. Landsat uses six scientific bands, organized through the existing input adapter. MODIS supplies seven bands on a native 5 by 5 grid, represented as 25 coarse tokens per time step. The optical training footprint spans approximately 2,560 projected metres; the MODIS grid spans approximately 2,500 projected metres. MODIS is not upsampled to simulate fine spatial detail. The current experiments do not use Sentinel-1, although an archive audit found all 12 monthly Sentinel-1 records for 1,027 of the regions.

Auxiliary supervision comprises shoreline labels from an existing JRC 2020–2024 aggregation, GLC-FCS30D land-cover labels from 2022, a Copernicus GLO30 static surface-elevation product, and tide-model-derived monthly targets. The elevation product is a digital surface model, not underwater bathymetry, and its edition year does not imply acquisition in 2024. Spatial targets are aligned to a common origin and a 100 by 100 grid with 25.6 m spacing. Differences in source dates mean that these targets are not fully synchronous observations.

The regions span approximately 47.08 degrees south to 51.93 degrees north. Europe and western Asia contribute 275 regions, North America 245, Africa and the Middle East 232, South America 149, Oceania 146, East Asia and the northern Pacific 57, and southern and southeastern Asia 46. Thus, coverage is geographically broad but uneven. These regions have not yet undergone a spatial-overlap audit against downstream datasets or the foundation models' pretraining corpora.

| Water fraction within valid JRC pixels | All regions | Training regions |
|---|---|---|
| 0% | 150 | 121 |
| More than 0%, at most 25% | 309 | 246 |
| More than 25%, less than 75% | 320 | 250 |
| At least 75%, less than 100% | 232 | 178 |
| 100% | 139 | 116 |
| Total | 1,150 | 911 |

Table 1. Composition of aligned coastal labels. Fractions describe valid observed pixels, not the entire unmasked tile. All-water regions are not necessarily exclusively marine. Original sampling-footprint fractions differ from these cropped-label statistics.

### 2.2. Downstream tasks

We evaluate MagicBathyNet bathymetry regression, LICS shoreline segmentation, MARIDA marine surface segmentation, and MADOS marine surface segmentation. Bathymetry is measured by root mean squared error (RMSE) in metres; segmentation is measured by mean intersection over union (mIoU). Dataset names identify the local benchmark inputs; our adaptations and splits should not be interpreted as a universal reproduction of published dataset leaderboards.

| Benchmark | Train | Validation | Test | Metric |
|---|---|---|---|---|
| MagicBathyNet | 2,030 | 226 | 566 | RMSE, lower is better |
| LICS | 80 | 20 | 100 | mIoU, higher is better |
| MARIDA | 694 | 328 | 359 | mIoU, higher is better |
| MADOS | 2,657 | 1,186 | 1,311 | mIoU, higher is better |

Table 2. Sample counts in the local evaluation manifests. LICS uses the first 80 samples of the local fine-tuning partition for training and the remaining 20 for validation. MagicBathyNet uses the existing stored split IDs; MARIDA and MADOS use the partitions exposed by their local loaders. Canonical dataset citations, acquisition details, and release identifiers must accompany a public release of these manifests.

### 2.3. Linear probing

We freeze the feature extractor and train a fresh linear probe for each downstream task. Following the OLMoEarth evaluation design [1], each trial runs for 50 epochs. The learning-rate candidates are 0.0001, 0.0005, 0.001, 0.005, 0.01, 0.05, 0.1, and 0.5. AdamW uses weight decay 0.01, batch size 32, and five epochs of linear warmup followed by cosine decay towards 0.00001. Probe initialization uses seed 42. Validation selects the configuration after each 50-epoch trial; the selected configuration is then evaluated on the test split. Test scores do not select the learning rate or normalization.

For OLMoEarth, the input patch size is four. We compare the official pretrained normalization statistics with benchmark statistics using the same mean-and-standard-deviation scaling. Most benchmark statistics are fitted on the training inputs; MADOS uses published statistics embedded in the existing evaluation implementation. The current tasks use single-time-step inputs. External models retain their native adapters and input-size requirements, so identical optimization budgets do not imply identical sensor information or preprocessing.

For segmentation, a linear layer maps each spatial feature to class logits for its corresponding output patch; logits are resized to the original target grid when required. Cross-entropy ignores label 255 and does not use additional class weights. For regression, a single-channel 1 by 1 convolution predicts bathymetry and minimizes mean squared error over valid pixels. Neither task uses the coastal multitask loss during probing. Features are extracted with BF16 autocast and stored in float16 caches in the current implementation.

## 3. Optical–depth dual-foundation method and controls

### 3.1. Official checkpoint references

We evaluate official OLMoEarth Nano and Large checkpoints alongside TerraMind, Prithvi EO-2.0, DOFA, SatMAE, CROMA, Scale-MAE, AnySat, and TESSERA. OLMoEarth Nano and Large emit 128- and 1,024-dimensional features, respectively. Several interfaces require sensor or band adaptation. Scale-MAE uses RGB, whereas the other adapters can use different spectral subsets. The present TESSERA experiment

supplies a single optical observation with the existing temporal-bin adaptation and does not evaluate its full annual multimodal setting. Lower scores under these interfaces should not be taken as upper bounds on the underlying models' capabilities.

## 3.2. OLMoEarth–PromptDA feature fusion

The primary method combines the official OLMoEarth Large optical encoder with the official PromptDA ViT-L depth foundation model [3]. OLMoEarth receives the optical bands through its native sensor adapter. PromptDA receives an RGB rendering of the same observation through its native image preprocessing. The two encoders are evaluated independently and remain frozen throughout feature extraction and probing.

The OLMoEarth input is 256 by 256 with patch size four in the probing protocol. PromptDA receives 448 by 448 RGB inputs with its native patch size fourteen. Its 32 by 32 final normalized patch-token grid is bilinearly aligned with the OLMoEarth spatial grid. We concatenate the aligned channels:

$$F_{dual} = [F_{OLMoEarth};\ \text{Resize}\ (F_{PromptDA})] \in \mathbb{R}^{H \times W \times 2048}.$$

PromptDA contributes a representation learned for depth estimation; the fusion does not use predicted depth, ground-truth depth, a depth prompt, or a learned projection. The same two encoder weights are used for all four tasks. A fresh linear probe is trained on the concatenated features, so the reported dual-foundation result measures transfer of a task-independent representation. The concatenated probe has twice the feature width of the OLMoEarth-only probe, an acknowledged source of additional capacity.

## 3.3. Coastal adaptation of Nano as a control

The no-MoE baseline retains the official Nano transformer and adds a native MODIS input adapter and coastal task heads. The training objective is

$$L_{coast} = L_{shore} + 1.6L_{landcover} + 0.1L_{elevation} + 0.5L_{tide}.$$

Shoreline uses binary cross-entropy, land cover uses cross-entropy, and elevation and tide use SmoothL1. Elevation targets use a signed logarithmic transformation in the coastal training implementation. Training runs for 40 epochs with batch size four and seed 42. The official parameters are frozen for the first three epochs, then updated at learning rate 0.000045; added modules use 0.0045. AdamW weight decay is 0.05. A separate coastal validation objective selects the checkpoint; it uses a 0.1 coefficient for mean tide and is not identical to the training objective.

The Nano MoE variants use six experts with corrected Top-1 or Top-2 routing and physical token coordinates. Coordinates encode token centres using each modality's pixel spacing and origin before common normalization. This changes relative spatial representation; it does not supply a new global location signal. Corrected Top-1 uses a straight-through gate because softmax over a single selected logit is constant. Load-balancing and expert-diversity penalties have coefficients 0.01 and 0.001, with a three-epoch diversity warmup.

The unfrozen Top-1 and Top-2 runs use the coastal validation-selected epochs 39 and 30. In a separate experiment, the official Nano parameters remain frozen throughout training while added modules and heads are updated. These runs use seed 43 and BF16; we report their fixed epoch-40 snapshots. Consequently, frozen versus unfrozen comparisons also differ in seed, precision, and checkpoint selection and are not isolated causal tests of freezing. Coastal training uses patch size eight, whereas downstream probing uses patch size four. Downstream inputs do not include paired MODIS, and the coastal task heads are discarded.

### 3.4. Target-task-adapted SpectralMoE as a secondary control

We also adapt the released SpectralMoE architecture of Chen et al. [2], replacing its visual foundation model with OLMoEarth Large while retaining the PromptDA encoder and the released six-expert, Top-1 adapter design. This is an implementation adaptation of prior work, not a newly introduced dual-gated MoE architecture. Visual inputs use 256 by 256 with patch eight, and PromptDA uses 448 by 448 with patch fourteen. The interface aligns the spatial grids and averages the visual spectral groups as implemented.

Each downstream task separately trains MoE and task-head parameters for 20 epochs with the two foundation encoders frozen. Segmentation uses the adapted Mask2Former head; bathymetry uses a regression head and MSE. We then discard the task head, freeze the adapted encoder and MoE, and train a new linear probe. The representation has therefore already received target-task supervision before probing. The adapter weights differ by task, and these scores cannot be interpreted as those of a single task-independent foundation model. Interface adaptations and the retained author implementation also prevent claiming an exact reproduction of SpectralMoE's original benchmark recipe.

## 4. Results

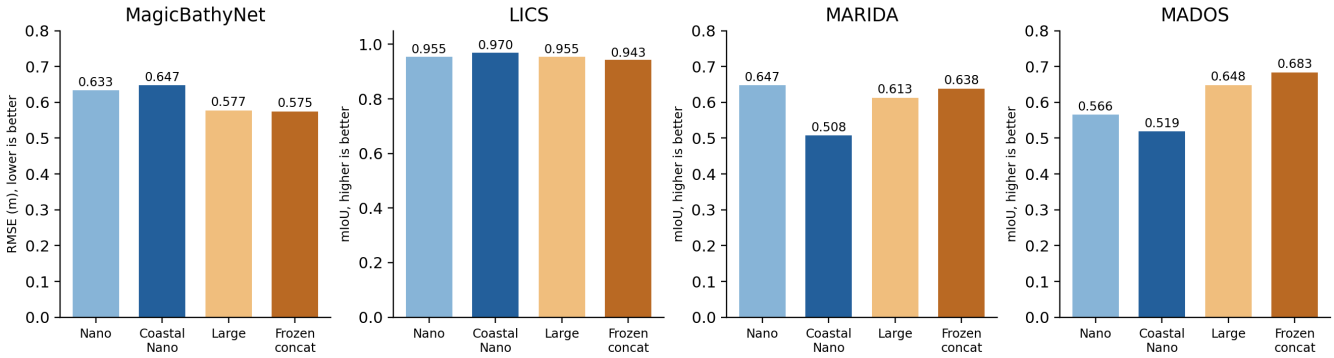



Figure 1. Task profiles of official Nano, coastally adapted Nano without MoE, official OLMoEarth Large, and the frozen OLMoEarth–PromptDA dual-foundation representation. Values are final test metrics under the local linear-probing protocol; lower is better for RMSE and higher for mIoU. Bars start at zero. Each bar represents one run, without uncertainty estimates.

### 4.1. Official models provide different reference points

No official checkpoint leads all four local tasks (Table 3). TerraMind achieves the lowest bathymetry RMSE and highest MADOS mIoU in this group. Prithvi leads LICS, whereas Nano leads MARIDA. The contrast between Nano and Large on MARIDA cautions against treating encoder size as a reliable predictor of performance under the current sensor interfaces and probing configuration.

| Official checkpoint | Magic RMSE | LICS mIoU | MARIDA mIoU | MADOS mIoU |
|---|---|---|---|---|
| OLMoEarth Nano | 0.633355 | 0.955118 | **0.647160** | 0.565515 |
| OLMoEarth Large | 0.577098 | 0.954755 | 0.612900 | 0.647969 |
| TerraMind Base | **0.529318** | 0.959422 | 0.499219 | **0.716325** |
| Prithvi EO-2.0 300M | 0.605855 | **0.968702** | 0.493729 | 0.641364 |
| DOFA ViT-Base | 0.677503 | 0.911565 | 0.413663 | 0.549592 |
| SatMAE ViT-Base multispectral | 0.621436 | 0.955805 | 0.374972 | 0.617544 |
| CROMA Base | 1.397860 | 0.922307 | 0.198291 | 0.212326 |
| Scale-MAE ViT-Large | 0.664649 | 0.709480 | 0.174337 | 0.333993 |
| AnySat Base | 0.556970 | 0.442266 | 0.239017 | 0.222442 |
| TESSERA v2 Medium | 0.698643 | 0.932338 | 0.365159 | 0.500293 |

Table 3. Official checkpoints under local frozen probing. Bold marks the best value in each column. Input adaptations differ; these are local evaluation results rather than official reported model scores.

## 4.2. Coastal adaptation changes the task profile

Coastal supervision improves the no-MoE Nano baseline on LICS by 1.46 percentage points but reduces its MARIDA and MADOS scores by 13.96 and 4.66 points, respectively (Table 4). Its bathymetry RMSE also increases. The combination of a shoreline gain and marine-task losses is consistent with specialization, but does not uniquely establish catastrophic forgetting: altered input handling, checkpoint selection, and patch-size transfer are alternative contributors.

Keeping the official encoder frozen changes this trade-off. Fixed epoch-40 Top-1 adaptation yields bathymetry RMSE of 0.568042 m, a 10.31% reduction relative to the official Nano reference, while remaining below that reference on all three segmentation tasks. Top-2 does not consistently dominate Top-1. We therefore report these variants as different task profiles, rather than identifying one as a generally better coastal foundation model.

| Nano configuration | Magic RMSE | LICS mIoU | MARIDA mIoU | MADOS mIoU |
|---|---|---|---|---|
| Official Nano | 0.633355 | 0.955118 | **0.647160** | **0.565515** |
| Coastal adaptation, no MoE | 0.647476 | **0.969692** | 0.507578 | 0.518878 |
| Unfrozen, physical coordinates, Top-1 | 0.633957 | 0.961896 | 0.486897 | 0.515103 |
| Unfrozen, physical coordinates, Top-2 | 0.632331 | 0.962021 | 0.491778 | 0.522514 |
| Frozen, physical coordinates, Top-1, epoch 40 | 0.568042 | 0.935636 | 0.540643 | 0.492267 |
| Frozen, physical coordinates, Top-2, epoch 40 | **0.564655** | 0.925591 | 0.537782 | 0.498283 |

Table 4. Coastal adaptation and transfer. Frozen/unfrozen refers to the official encoder during coastal training; all feature extractors are frozen during downstream probing. The no-MoE encoder contains approximately 1.700 million parameters and the Nano MoE encoders approximately 2.274 million, excluding downstream heads.

### 4.3. Optical and depth foundation features are complementary on two marine tasks

The frozen OLMoEarth–PromptDA dual-foundation representation improves MARIDA by 2.55 percentage points and MADOS by 3.54 points relative to OLMoEarth Large alone (Table 5). Bathymetry RMSE changes from 0.577098 to 0.574892 m, while LICS decreases by 1.16 points. The dual-foundation score remains below the best reference in the full model survey on each task, including Nano on MARIDA and TerraMind on MADOS. The result nevertheless supports the specific method claim: features from a depth foundation model can complement an optical foundation model for selected marine transfers without target-task adaptation of either encoder.

| Large-based configuration | Before probing | Magic RMSE | LICS mIoU | MARIDA mIoU | MADOS mIoU |
|---|---|---|---|---|---|
| Official OLMoEarth Large | No adaptation | 0.577098 | 0.954755 | 0.612900 | 0.647969 |
| OLMoEarth Large + PromptDA depth foundation | No adaptation | 0.574892 | 0.943147 | 0.638368 | 0.683345 |
| Large + PromptDA + SpectralMoE | 20 epochs on each target task | 0.560616 | 0.954343 | 0.495718 | 0.594606 |

Table 5. OLMoEarth-only, frozen optical–depth fusion, and target-task-adapted SpectralMoE settings. The last row has additional target-task supervision and uses patch eight rather than patch four. It is not a controlled test of MoE versus the dual-foundation method. The frozen dual-foundation combination contains approximately 612.139 million parameters; the task-adapted MoE combination contains approximately 617.877 million, excluding unused depth prediction and downstream heads.

The task-adapted MoE configuration does not recover the MARIDA or MADOS performance of the frozen optical–depth fusion, despite having received additional labels before probing. Differences in patch size, interface, head, and optimization preclude a causal claim that MoE is inferior. They do show why a dual-foundation method should first be compared with each frozen encoder and why target-task adaptation should not be conflated with complementary pretrained features.

## 5. Discussion: preserving representations while learning coastal corrections

Together, the experiments support a focused interpretation of the dual-foundation method. OLMoEarth supplies optical and multispectral evidence, whereas PromptDA supplies a depth-oriented visual representation. Their frozen combination improves two marine-surface transfers, but the unchanged encoders do not guarantee gains on every task. The method therefore offers a low-supervision baseline for testing whether geometric priors learned outside remote sensing can complement optical Earth-observation features.

A concrete next experiment is to learn a gated correction on top of the dual-foundation representation while preserving both original feature paths:

$$F_{out} = F_0 + \alpha(x)\,\Delta(F_0, F_{aux}).$$

Here, both pretrained encoders can remain frozen, the residual output can start at zero, and a feature-preservation penalty can discourage unnecessary departures from the two pretrained representations. This is a proposed experiment, not a method validated in this paper, and residual gating and distillation are established design ideas rather than claimed inventions. Its testable question is whether coastal supervision can improve the dual-foundation representation without removing the MARIDA and MADOS gains

observed under frozen probing. The minimum controls are OLMoEarth alone, PromptDA alone, frozen concatenation, a single-encoder adapter and a two-encoder adapter with matched training data and search budgets.

Several boundaries affect interpretation. First, the reported configurations are single runs without confidence intervals; small differences, especially the 0.38% bathymetry improvement from concatenation, should not be treated as established gains. Second, changes in data normalization, sensor mapping, and feature dimensions complicate cross-model ranking. Third, the coastal subset is geographically imbalanced and spatial overlap with evaluation data has not been audited. Fourth, PromptDA contributes features learned in a different domain; neither the use of its encoder nor improved scores proves that metric depth is the relevant information. Finally, four optical tasks do not cover sea-surface temperature, chlorophyll, sediment concentration, SAR performance, or full-year temporal reasoning.

These boundaries suggest focused follow-up tests. A dimension-matched fusion control would test whether extra probe capacity explains part of the observed benefit. A learned projection or low-rank fusion would test whether channel concatenation is sufficient. Matched-seed and fixed-patch comparisons would separate preprocessing from encoder complementarity. Paired Sentinel-1 inputs could then assess whether the optical–depth combination remains useful when a second remote-sensing modality is added. None of these outcomes is assumed by the present study.

## 6. Conclusion

In the evaluated coastal setting, combining OLMoEarth with a frozen depth foundation model provides complementary features for MARIDA and MADOS under a common linear-probing protocol. The same dual-foundation representation does not improve every task, and its larger probe and interface differences limit causal attribution. The result establishes a concrete baseline for optical–depth fusion in coastal remote sensing and motivates future learned fusion that preserves both pretrained feature paths.

## Data, code, and manuscript status

The project repository is https://github.com/NNNoreply/Coastal_Foundation_Models. Its initial release contains a project overview; code, model checkpoints, and data are not yet included. This working draft is based on existing local experiment records. Author names, affiliations, contact details, dataset permissions, and complete bibliographic metadata remain to be supplied before submission. Existing model and dataset creators retain attribution for their resources. AI assistance was used to organize and draft this manuscript from local code and result records; author verification of the submitted text and interpretations remains pending.

## Source references

The following are source pointers verified in the local repository documentation, not a completed bibliography. Canonical citations for every benchmark, model, geospatial product, optimizer, and proposed residual/distillation context remain to be added and checked before public submission.


1. OLMoEarth project. Official model and evaluation implementation. https://github.com/allenai/olmoearth_pretrain. The repository links the original report at https://arxiv.org/abs/2511.13655. Nano and Large in this study use different released model versions.
2. Xi Chen, Maojun Zhang, Yu Liu, and Shen Yan. *Local Precise Refinement: A Dual-Gated Mixture-of-Experts for Enhancing Foundation Model Generalization against Spectral Shifts*. arXiv:2603.13352, 2026. https://arxiv.org/abs/2603.13352. Implementation:

https://github.com/daxichen/SpectralMoE-main. Author/title metadata follow the released repository citation.
3. Haotong Lin et al. *Prompting Depth Anything for 4K Resolution Accurate Metric Depth Estimation*. arXiv, 2024. https://promptda.github.io/. The experiment uses the released PromptDA ViT-L visual encoder checkpoint.
4. *CROMA: Remote Sensing Representations with Contrastive Radar-Optical Masked Autoencoders*. Paper linked in the local official implementation: https://arxiv.org/abs/2311.00566.
5. SatMAE project and paper, linked in the local official implementation: https://sustainlab-group.github.io/SatMAE/ and https://arxiv.org/abs/2207.08051.
6. *Scale-MAE: A Scale-Aware Masked Autoencoder for Multiscale Geospatial Representation Learning*. Paper linked in the local implementation: https://arxiv.org/abs/2212.14532.